How to use Command Line Interface (CLI)

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Global Master of Business Administration

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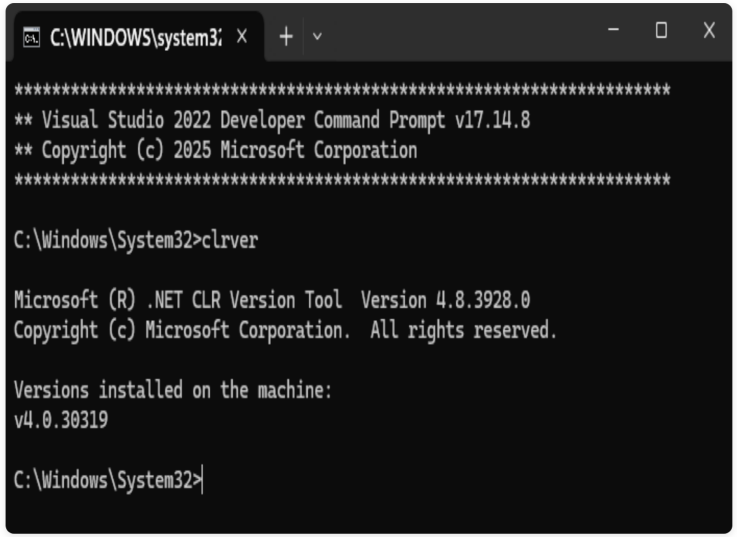
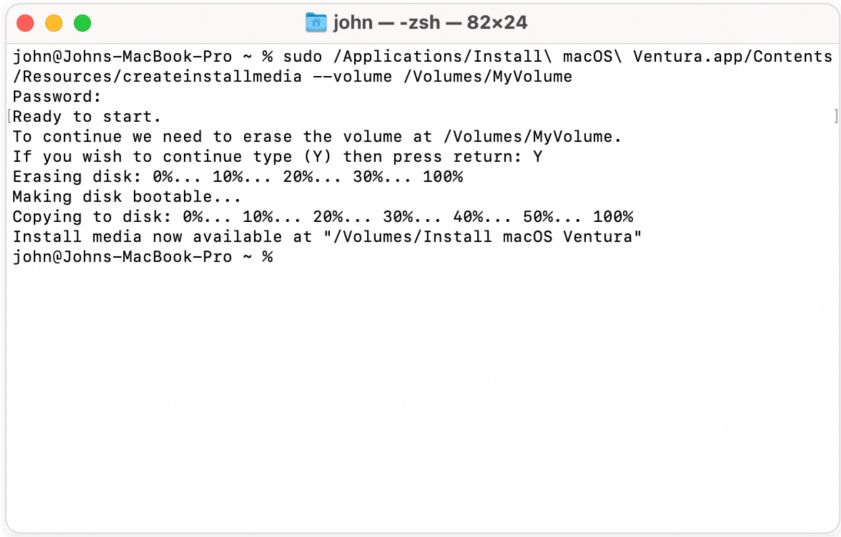
1. Purpose

This tutorial introduces the **command line interface (CLI)**, a way to control your computer by typing text commands, rather than clicking icons, on macOS/Linux (Unix) and Windows. Proficiency in the CLI is a foundational skill for business and non-engineering students enrolled in artificial intelligence and machine learning courses.





2. Essential Terminology (TUI Glossary)

Directory (Folder)	A location in a file system used to store and organize files and other directories.
Parent Directory	A directory that contains subdirectories. E.g., `climate_research` is the parent of `src`.
Child Directory	A directory inside another. E.g., `src` is a child of `climate_research`.
Current Working Dir	The directory you are currently "inside" in the terminal.
Home Directory	Default personal directory (~ on macOS/Linux, `C:\Users\Username` on Windows).
Root Directory	Top-most directory (`/` on macOS/Linux, `C:\` on Windows).
Project Root Dir	The top-most directory specific to a project, containing all related files and subdirectories.
Absolute Path	Complete path from the root (e.g., `/home/user/climate_research`).
Relative Path	Path relative to current dir (e.g., `src/main.py` or `../references`).
Terminal / Shell	The text-based application (TUI) used to type and execute commands.

3. How to Open the Terminal / Command Prompt

macOS Terminal

- **Spotlight**: `Cmd + Space` → type `Terminal` → `Enter`
- **Applications**: Finder → `Applications` → `Utilities` → Terminal

Windows Command Prompt

- **Start Menu**: `Win key` → type `cmd` → `Enter`
- **Run Dialog**: `Win + R` → type `cmd` → `Enter`

Linux Terminal

- **Keyboard Shortcut**: `Ctrl + Alt + T` (Ubuntu & most distros)
- **Application Launcher**: Search for `Terminal` or `Console` in app menu

4. Common Commands: Navigation

Action	macOS / Linux	Windows (Cmd)
List files	`ls` (or `ls -la`)	`dir`
Print working dir	`pwd`	`cd`
Move to directory	`cd directory_name`	`cd directory_name`
Move to parent	`cd ..`	`cd ..`

4. Common Commands: Managing Directories

Action	macOS / Linux	Windows (Cmd)
Create directory	`mkdir directory_name`	`mkdir directory_name`
Create subdirectory	`mkdir -p parent/sub`	`mkdir parent\sub`
Copy directory	`cp -r src dest`	`xcopy /E /I src dest`
Delete directory	`rm -rf directory_name`	`rmdir /s directory_name`

4. Common Commands: Managing Files

Action	macOS / Linux	Windows (Cmd)
Create text file	`touch file.txt`	`type nul > file.txt`
Open in GUI editor	`touch f && open -e f`	`notepad file.txt`
View file contents	`cat file.txt`	`type file.txt`
Rename	`mv old_name new_name`	`ren old_name new_name`
Delete file	`rm file.txt`	`del file.txt`

4. Common Commands: Directory Structure

Action	macOS / Linux	Windows (Cmd)
Display tree	`tree`	`tree /F`

Tip: On macOS/Linux, install via `brew install tree` or use `find .`. On Windows, `/F` shows files; without it, only folders are listed.

5. Detailed Examples: Setup & Navigation

Setup: Create a practice directory and navigate into it:

```
mkdir terminal_practice
cd terminal_practice
```

List Files & Directories

```
# macOS/Linux
ls
ls -la

# Windows
dir
```

Interpreting output: `d` = directory, `-` = file, `l` = symlink. On Windows: `<DIR>` = directory, number = file.

Print Working Dir & Navigate

```
# Print current location
pwd       # macOS/Linux
cd        # Windows

# Move into a directory
mkdir documentation
cd documentation

# Move to parent directory
cd ..
```

5. Detailed Examples: Managing Directories

Create a directory:

```
mkdir src
```

Verify with `ls` or `dir`.

Create nested subdirectories:

```
# macOS/Linux
mkdir -p paper/drafts

# Windows
mkdir paper\drafts
```

Copy a directory:

```
# macOS/Linux
cp -r src src_backup

# Windows
xcopy /E /I src src_backup
```

Rename a directory (`documentation` → `docs`):

```
# macOS/Linux
mv documentation docs

# Windows
ren documentation docs
```

5. Detailed Examples: Managing Files

Create Empty File

```
# macOS/Linux
touch notes.txt

# Windows
type nul > notes.txt
```

Open in GUI Editor

```
# macOS
touch notes.txt && open -e notes.txt

# Windows
notepad notes.txt
```

View File Contents

```
# macOS/Linux
cat notes.txt

# Windows
type notes.txt
```

Delete a File

```
# macOS/Linux
rm notes.txt

# Windows
del notes.txt
```

Visualizing the Directory Structure

```
# macOS/Linux
tree

# Windows
tree /F
```

On macOS, install via `brew install tree` or use `find .`. On Windows, `/F` shows files.

6. Student Evaluation: Scenario

Setting up a Research Project Workspace: The practice root directory and all subdirectories must include your English name (e.g., `james`):

Type	Example Name
Project root directory	`climate_research_james`
Subdirectories	`src_james`, `output_james`, `paper_james/drafts_james`, `references_james`, `doc_james`
File in doc	`todo.txt`: first line is your name, then task list

6. Step-by-Step Practical Assessment

Step 1: Open the Terminal / Command Prompt.

Step 2: Initialize Create `climate_research_[yourname]`, move into it.

Step 3: Structure Create `src_`, `output_`, `paper_/drafts_`, `references_`, `doc_` subdirectories.

Step 4: Files Create `doc_[name]/todo.txt` (name + task list), `references_[name]/sources.txt`.

Step 5: Rename `doc_[name]` → `documentation_[name]`, `sources.txt` → `citations.txt`.

Step 6: Cleanup Navigate to root, delete `output_[name]`.

Step 7: Verify While inside `climate_research_[yourname]`, run:

```
# macOS/Linux
date && tree && cat documentation_[yourname]/todo.txt

# Windows (Cmd)
echo %date% %time% && tree /F && type documentation_[yourname]\todo.txt
```

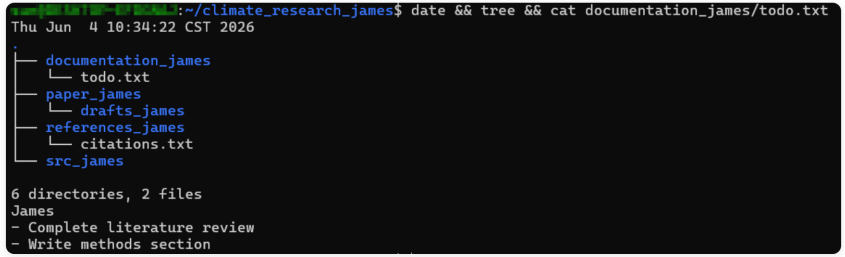
Step 8: Exit Navigate back to the parent directory with `cd ..`

6.3 Deliverables

Directory Structure Screenshot

Take a screenshot of the entire terminal window showing:

- Execution of the verification command (`date && tree` or Windows equivalent)
- Your custom root directory name `climate_research_[yourname]`
- All named subdirectories
- The printed timestamp



Submit the screenshot to **iLearn** for grading.

6.4 Evaluation Criteria

- ☐ **Terminal Launch** : Student can open the terminal or command prompt correctly.
- ☐ **Navigation** : Student can move between nested directories without getting lost (`cd` , `cd ..`).
- ☐ **Pathing** : Student understands how to create nested directories with custom naming (e.g., `mkdir -p` or Windows equivalent).
- ☐ **File Management** : Student is able to create files inside specific directories and rename directories/files correctly.
- ☐ **Safety** : Student can safely delete files and empty or non-empty directories.
- ☐ **Verification** : Student is able to filter (grep) their terminal history to generate the verification report.